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Computers in Paper Machine Clothing

There has been a computer revolution within the last two decades that has changed every aspect of life drastically. One of the profound effects of this change has been in the area of product and process design, development and manufacturing. Like in every other industry, today computers are widely used in every stage of papermaking and paper machine clothing from product concept design to shipping and receiving.

Parallel to the expansion of computer application areas in industry, the computer technology itself went through impressive developments within the last two decades. The development of computer technology, both software and hardware, is ever increasing without any pause in the foreseeable future. Computers are getting faster and more powerful in terms of memory, versatility, adaptability, performance and supporting software programs. The change is so fast that any specific description of the current computer hardware and software may be outdated in a few years or even months. Therefore, the purpose of this chapter is not to discuss the computer technology but to demonstrate the current and potential application areas of computers.

Probably the most exciting development that is taking place last few years is in the information area. Using World Wide Web (WWW) on the Internet, users can connect to any other computer in the world (provided that it is also connected to the net), browse a

web page, send e-mail, participate in discussions and video conferences, etc. People use this technology to exchange ideas, solve problems and discuss issues. Federal and state institutions, private and public companies and individuals are developing their own homepages full of information about themselves, their products, companies and services. Homepages can be designed in any format by the programmer. Figure 8.1 shows a homepage developed for the pulp and paper industry.

By pointing the cursor and clicking any item on the homepage, one can go into a specific topic for more information.

8.1 Definitions

In order to fully understand this chapter, a glossary of terms follows.

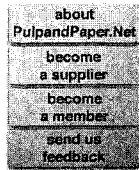
- **account (Computer Account)**—The name used to uniquely identify users of a computer resource. Referred to variously as GID (global ID), computing account, userid or simply “ID.”
- **assembly language**—This level of language is between high level language and machine language. The commands are in English letters but they may not have any literal meaning since they are abbreviated. Execution of assembly language programs are

Welcome to Pulp and Paper.Net

An Easier Way to Get
More Information
in Less Time.



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Suppliers and Products

- ◊ Browse Segments
- ◊ Search for Company or Product

Classified Ads

- ◊ Job Opportunities
- ◊ Used Equipment - Coming soon

Trade Shows

- ◊ Tradeshow Search
- ◊ Tradeshow Scrapbook

From the 'Net

- ◊ Pulp and Paper Jumplist
- ◊ Newsgroup Topics

Business News

- ◊ Walden Paper Report
- ◊ Walden Fiber and Board Report
- ◊ The Green Paper/Fortrans

- ◊ You must register to access these areas.◊

NEW Pulp and Paper.Net Welcomes Voith Sulzer ◊

Suppliers and Products

This section is where you can easily find information on companies, products, chemicals or services that serve the pulp and paper industry. Whether you know exactly what you are looking for or just want to browse, our navigation system brings you the information you are seeking.

- ◊ Browse Market Segments
- ◊ Search for Company or Product

TradeShows

Pulp and paper industry tradeshow are another valuable resource for industry information. Pulp and Paper.Net wants to keep you informed about tradeshow and seminars that may be of interest to you. This section offers you information on upcoming tradeshow and seminars and includes images from past tradeshow we have attended.

- ◊ Tradeshow Search
- ◊ Tradeshow Scrapbook

NEW Business News

In the information age, staying up to date on industry information is not an easy task. This section provides pulp and paper industry information. Whether it is unique information we bring to you or other pulp and paper on-line resources, we want to help you find the information quicker and easier.

- ◊ Walden Paper Report
- ◊ Walden Fiber and Board Report
- ◊ The Green Paper/Fortrans

Classified Ads

Whether you are in the market for a new job or just curious about available openings, this section of Pulp and Paper.Net allows you to search the Job Opportunities database. If you locate a posting that interests you, Pulp and Paper.Net will provide the information you need to confidentially contact the company posting the job opportunity.

- ◊ Job Opportunities
- ◊ Coming soon - Used Equipment

From The 'NET

The Internet is a valuable resource for information, but finding that information is not always easy. Pulp and Paper.Net's goal is to make pulp and paper producers' jobs easier by putting them closer to the information they need. This section of Pulp and Paper.Net provides information and links to other sites on the Internet containing industry information. Make a bookmark for Pulp and Paper.Net and begin your search for pulp and paper industry information with us.

- ◊ "The Pulp and Paper Jumplist"
- ◊ Newsgroup Topics



FIGURE 8.1. A homepage for the pulp and paper industry (<http://www.pulpandpaper.net/>).

- faster than high level language programs because there is no need for compilation. That is why video games, for which speed is critical, are written in assembly language. Assembly language is more difficult to write but gives more control of computer to the user.
- bitnet—a worldwide computer network
 - browser (Web Browser)—It is a software program that provides a graphical user interface to the Internet's World-Wide Web (WWW). There are several browser programs currently in use including Netscape and Mosaic.
 - CD (compact disk)—Digital disk for data storage.
 - CPU (central processor unit)—A microchip inside the computer that runs all the software and operations of the computer.
 - cyberspace—A generic name for the computer environment that includes e-mail, WWW and networks, etc.
 - desktop—A personal computer that can fit on top of a desk.
 - digitizer—A device that converts analog data, figures, etc., to digital form.
 - download—Transferring information or files from one computer to another. Downloading generally implies that the data are moving from a larger, more powerful machine to a smaller one.
 - DPI—dots per inch on the computer monitor.
 - EDI (electronic data interchange)—Transfer of data via computers from place to place.
 - e-mail (electronic mail)—Mail that is sent from computer to computer via telecommunications media.
 - expert system—computer software packages that are based on scientific data with empirical and theoretical knowledge of experts in the field. Expert systems are also called “intelligent systems.”
 - floppy disk—Portable computer disk for data storage.
 - FTP (file transfer protocol)—Used to transfer files from one computer to another over the Internet.
 - gopher—An on-line information retrieval system available on the Internet.
 - hard disk—High storage capacity permanent disk inside the computer
 - high level language—Programming languages such as Basic, FORTRAN, Pascal and ‘C’ that consist of special commands to write computer programs. The commands are in English language which is why it is called ‘high level’, i.e. level that a person can understand. A computer code written in English is converted to assembly language first (compilation) and then to machine language before it is executed by the computer. ‘C’ language has several versions and it has gained wide acceptance in recent years. Figure 8.2 shows comparison of high level language programs to calculate the area of a rectangle.
 - Homepage (or Page)—In the context of WWW, it refers to the entry point of a Web site. It is similar to a main menu which may consist of any number of screens full of information and hyperlinks to related items (Figures 8.1 and 8.3).
 - HTML (hyper text markup language)—HTML documents are text files that contain HTML tags to develop a home page. HTML tags tell a web browser (such as Netscape or Mosaic) how to format or display the text of the document. HTML tags are set off from the rest of the text by left and right angle brackets, i.e. <p> for paragraph tag. Although most HTML tags are paired, some are not. Paired tags have a beginning tag, such as <title>, and an ending tag such as </title>. Text affected by the tag is included between the paired tags (<title>My home page</title>). HTML is not case sensitive, i.e., <title> is the same as <Title> or <TITLE>. Tags that are not paired

PROGRAM IN FORTRAN

```

      PROGRAM    CALCULATING      AREA
      PRINT *, 'INPUT A, B'
      READ(*,*) A,B
      AREA = A*B
      WRITE (*,20) AREA
20    FORMAT (F10.2)
      STOP
      END

```

PROGRAM IN BASIC

```

10 print "enter values for a and b"
20 input a,b
30 let area = (a*b)
40 print "a=";a,"b=";b,"area=";area
60 end

```

PROGRAM IN C

```

#include <stdio.h>
#include <math.h>

main()
{

double a,b,area;
scanf("%lf %lf\n",&a,&b);
printf("a=%g,b=%g\n",a,b);
area=a*b;
printf("area=%g\n",area);

}

```

FIGURE 8.2. High level language program codes to calculate the area of a rectangle.

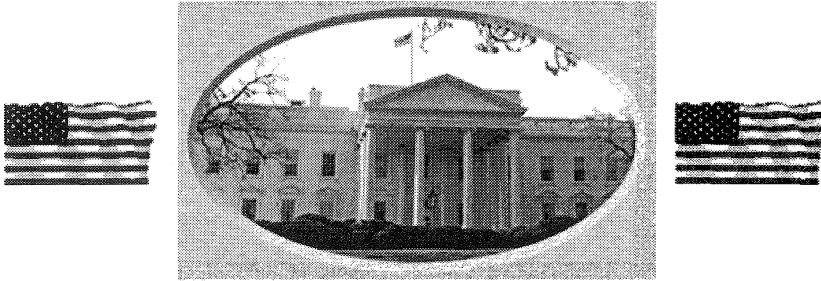
include paragraph tag <p>, the horizontal rule tag <hr>, and the line break tag
. There are several HTML editors and add-ons that can be used to develop a homepage. Figure 8.3 shows the homepage for the White House.

- Internet—A collection of networks and gateways around the world. Internet is the World's largest computer network

comprised of thousands of interconnected computer networks worldwide.

- IP addresses—A unique number that identifies each host computer or computer on a network. IP stands for Internet Protocol.
- laptop—A small personal computer that can fit on one's lap or briefcase.
- machine language—The commands in this language are made of 0's and 1's that computer can understand. Normally, programs are not written in machine language because of its difficulty. However machine language is very powerful compared to assembly and high level languages.
- main frame—A host computer that can run a large number of personal computers or workstations. Main frames are used in large corporations, universities and R&D centers.
- midi frame—A host computer smaller than main frames that is used in small to medium size companies to run personal computers and workstations
- mini frame—A host computer that can control and run several personal computers or workstations usually in an office.
- Mosaic—Mosaic is a web browser that provides a graphical user interface to the Internet's WWW.
- MS-DOS—Micro Soft Disk Operating System.
- Netscape—Netscape is a web browser that provides a graphical user interface to the Internet's WWW.
- network systems—These are communication systems that connect different information systems together all over the world. Examples of the worldwide network systems are Internet and Bitnet.
- Newsgroup—An on-line forum for discussion of related topics, accessible by a newsreader. Some newsgroups allow postings or messages from

Good Morning

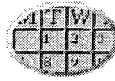


*Welcome to the
White House*



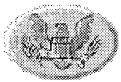
The President & Vice President:

Their accomplishments, their families, and how to send them electronic mail



What's New:

Remarks by the President: White House Commission on Aviation Safety and Security



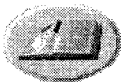
Interactive Citizens' Handbook:

Your guide to information and services from the Federal government



White House History and Tours:

Past Presidents and First Families, Art in the President's House and Tours



The Virtual Library:

Search White House documents, listen to speeches, and view photos



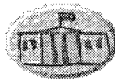
The Briefing Room:

Today's releases, hot topics, and the latest Federal statistics



White House Help Desk:

Frequently asked questions and answers about our service



White House for Kids:

Helping young people become more active and informed citizens



*Please sign the White House electronic guest book.
Or comment on this service at feedback@www.whitehouse.gov.*

FIGURE 8.3. Homepage of the White House (<http://www.whitehouse.gov/WH/welcome.html>)

anyone, while others may be moderated, i.e. postings are screened.

- Newsreader—Software that allows access to Usenet News and the newsgroups available therein.
- PC (personal computer)—A computer that has all the necessary operating system programs to run by itself, i.e. without connecting to a large computer. IBM (International Business Machines) and Apple Computer, Inc., are examples of the largest personal computer manufacturers in the world.
- PLC (programmable logical controller)—Device that is used to control and operate machines and equipment electronically
- Programming language—set of commands to instruct the CPU to perform a specific job. There are three levels of programming languages: high level language, assembly language and machine language.
- RAM (random access memory)—Memory that is used by CPU to run the software programs. RAM size should be larger than the total size of the programs that will be run. Turning the computer off will erase everything in the RAM.
- ROM (read only memory)—A permanent memory location in the computer that has the necessary commands to run the computer. ROM can not be accessed by the user and therefore can not be erased.
- scanning—digitizing a picture or text and loading it to the computer screen (i.e., opposite to printing).
- source (source code)—Source is the HTML codes for a Web document.
- Sun—A computer or workstation manufactured by Sun Microsystems.
- super computers—Extremely fast and powerful mainframes that are used mainly for research purposes in large universities and institutions such as NASA.
- Telnet—Telnet software allows a user to login to a remote computer and use its resources.
- Unix—An operating system used by Sun and other companies. Unix is the most widely used multi-user general purpose operating system in the world.
- upload—Transferring information or files from one computer to another. Upload implies that the data is moving from a smaller machine to a larger, more powerful one.
- URL (universal resource location)—The electronic address for an information source on the WWW or Gopher, such as an FTP site, gopher service or web page. For example, the URL for the Auburn University home page is <http://www.auburn.edu> and the American Forest and Paper Association is <http://www.afandpa.org>. Each address has a prefix which tells the Browser what kind of resource it is viewing. FTP site will have the prefix <ftp://>, a Gopher site will have the prefix <gopher://> and an HTML file that is read from the hard drive or diskette will have the prefix <file://>.
- virus (computer virus)—A computer virus is a computer program that is developed with malicious intent. These viruses can have very bad effects on the programs in standalone computers or networks. For example, viruses can turn a file that is created with a wordprocessor into a garbage. Viruses can replicate themselves and spread from computer to computer via floppy disks or through file downloading in a network. There are computer programs developed to scan computers and disinfect them if they are hit by a virus. There are many types of viruses and different disinfectant programs. Probably the best protection from viruses is to frequently make clean (i.e. free from virus) backup copies of working files.
- work station—A computer that is connected to a large central computer. A work station usually does not have a

CPU of its own. However, computers with CPUs can be used as workstations as well.

- WWW (World Wide Web)—A client-server hypertext distributed information retrieval system available over the internet. WWW was developed by CERN High-Energy Physics laboratories in Geneva, Switzerland.
- zip drive—A removable hard disk capable of holding anywhere from 100 MB (megabyte) to 1 GB (gigabyte) of data on a removable cartridge depending on the model of the drive. They are typically slower than hard disks and magneto-optical drives, but faster than tapes and CD ROMs.

8.2 Application Areas of Computers

8.2.1 Computer Aided Design (CAD) and Modeling

Computers have been widely used in design, development, modification and analysis

of various products and processes. A typical CAD system consists of computer, monitor, scanner, plotter, printer and digitizer.

Paper machine clothing is relatively expensive. Design and development of a new fabric by trial and error can increase product development costs and delay the introduction of a product into the market. Computers can be used to develop a product with correct geometrical and mechanical properties such that potential performance of the product can be estimated and a decision about the product can be made before actual manufacturing trials. More realistic computer programs make the decision making process simpler and faster thus reducing the cycle time from conception of an idea to actually making the product whether it is a fabric or paper.

Product design is an important application of CAD. CAD programs are used to design and characterize polymers, fibers, yarns, fabrics, nonwovens and paper. Use of CAD shorten the development time of new, innovative products. Figure 8.4 shows a fabric structure developed with a CAD system.

Modeling of a product in computer involves geometric modeling and mechanistic model-

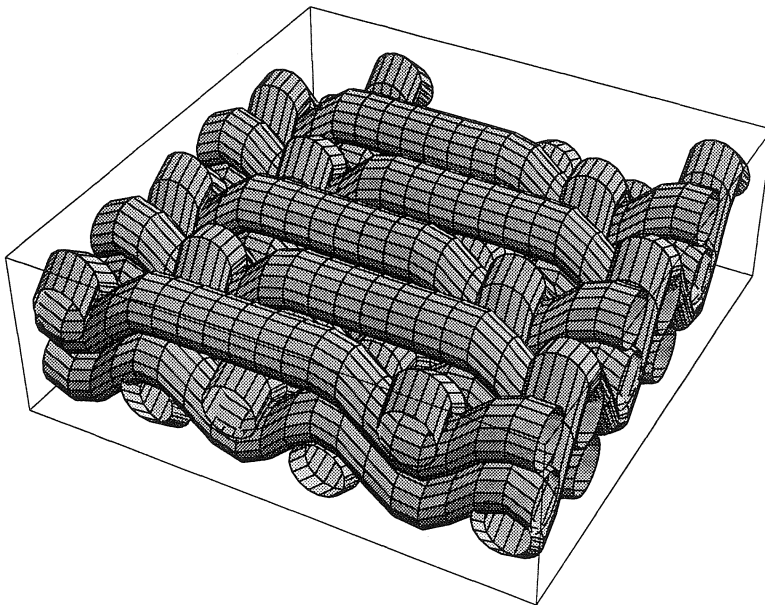


FIGURE 8.4. Dryer fabric structures developed using a CAD system.

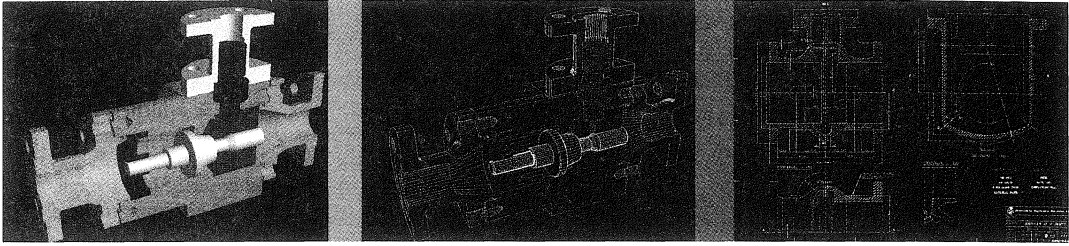


FIGURE 8.5. Steps in computer aided design of a product: drawing, 3D meshing and 3D modeling (photo courtesy of Research Engineers, Inc.).

ing. Realistic modeling of a product structure is critical for success. Therefore, modeling and computer aided design must be based on theoretical and empirical data available in the literature or through research. Generally, finite element analysis (FEA) method is used to model and analyze the structures. Paper machine clothing materials have nonlinear, visco-elastic properties which make modeling relatively more complicated. The major steps in computer aided design and modeling of a product are drawing, 3D meshing and 3D model generation as shown in Figure 8.5.

Various CAD software programs have been developed including AutoCAD®,

VersaCAD®, Algor®, MSC, etc. A good CAD system might be quite expensive and should be evaluated very carefully. The issues to consider when purchasing a CAD system are simplicity, customer support, availability of user groups, availability of manuals, software update options, flexibility and service.

8.2.2 Computer Integrated Manufacturing (CIM)

Computers are used in almost every manufacturing stage of paper machine clothing and

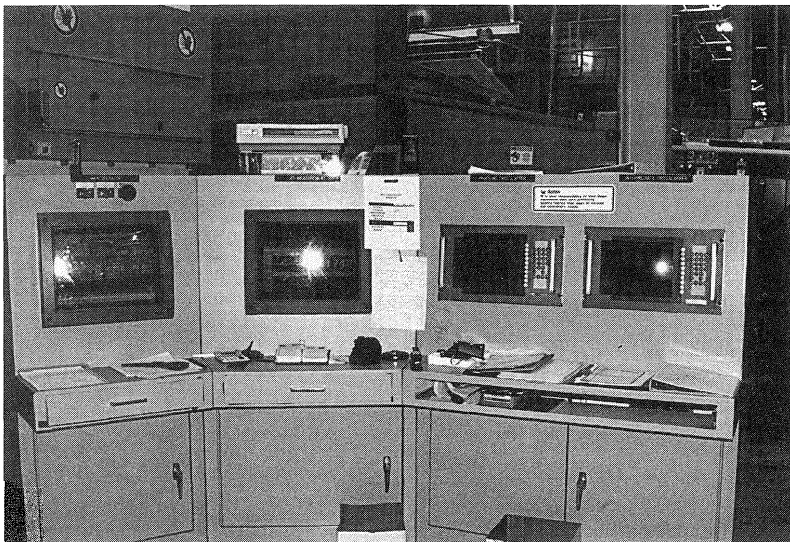


FIGURE 8.6. Computer control room for preneedling of press fabrics.

papermaking. Their usage includes testing of incoming raw materials, process control, property testing, etc. Most of the mechanisms of modern textile machines are electronically controlled and operated using programmable logical controllers (PLCs). With the PLCs, the performance of machines can be monitored by measuring picks per minute, stop and start times, etc. Figure 8.6 shows the computer control room for a preneedling machine for press fabric manufacturing.

CIM is an integrated computerized production system that is used to improve operations and product quality in a manufacturing environment. Benefits include savings in energy, labor, cost, maintenance and product quality improvements. Factory automation has been a major driving force behind CIM. CIM allows us to collect data in real time and eliminate waste and bottlenecks between processes by controlling each process while monitoring the data.

In general, CIM consists of three building blocks: process technology, control technology and knowledge technology. Process technology includes production machinery and automation to minimize manufacturing labor. Control technology uses computers, microprocessors and programmable controllers for self-regulatory manufacturing process control. Knowledge technology requires computers capable of interpreting information and decision making in real time. Decision science, expert systems, artificial intelligence and knowledge engineering are the tools to construct a CIM environment in a plant.

Production planning and control systems are used as part of Management Information Systems (MIS) to integrate planning, control and execution of manufacturing processes. These systems allow planning of quantities, checking delivery dates, monitoring order progress, materials management, capacity planning and scheduling. These are important elements for the success of Just in Time (JIT) and Quick Response (QR) systems. The purpose of JIT is to have the right product quantity at the right time and in the right place. QR shortens cycle (delivery) time of a product

and allows fast flow of information from supplier to the customer.

8.2.3 Process Simulation and Testing

Computers are widely used to simulate the process conditions either during manufacturing or during end use. For example, the behavior of a forming fabric during heatsetting can be predicted using computer simulation programs. Similarly, drainage characteristics of a forming fabric on the paper machine can be obtained using a fluid flow simulation program.

Computer testing of products such as fabrics and paper products can be done by first modeling the product and then subjecting it to the test conditions. For example, test simulation for tension, compression, bending, etc., of fabrics can be done using software packages that are specially developed for this purpose.

Today, most of the testing equipment are computerized for accuracy, reliability and speed. Figure 8.7 shows computerized tensile testing of yarns. For example, with this system, the modulus of yarn is calculated directly by the computer from the stress-strain curve which eliminates the human error and subjectivity.

8.2.4 Image Technology

High resolution computers are ideal tools for microscopic view of materials. A high resolution camera is connected to a computer for this purpose. Magnified views of products can be viewed on the computer monitor and printed for a hard copy (Figure 8.8). Imaging technology allows documents such as forms, drawings, pictures and charts to be scanned and digitized such that they can be manipulated electronically.

Computer imaging technology can be used to quantify various fabric properties such as void volume, surface smoothness and contact, etc.

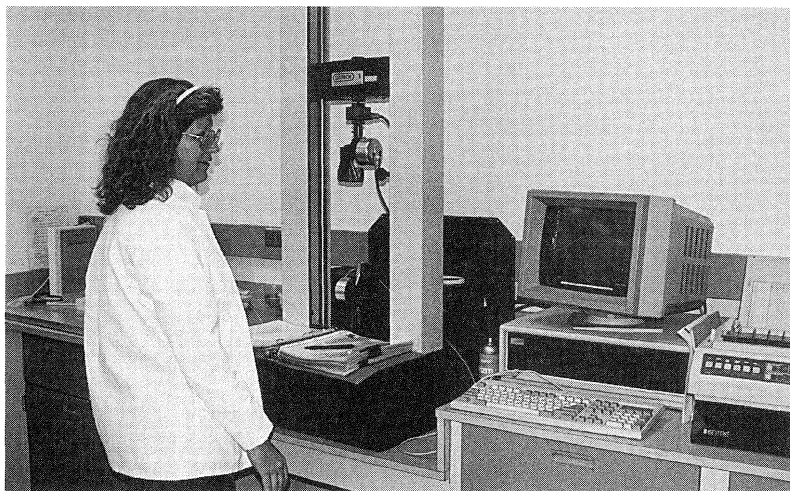


FIGURE 8.7. Computerized tensile testing.

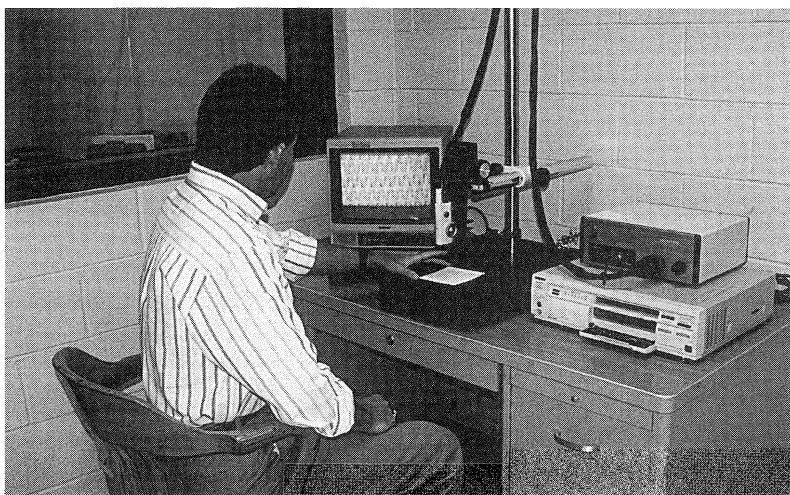


FIGURE 8.8. Image analysis of a dryer fabric.

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8.4 Review Questions

- 1 Discuss the effects of computer revolution on paper machine clothing and paper manufacturing.
- 2 What are the technical parameters to consider when designing a forming fabric using a CAD system in order to make it a realistic representation of the real fabric.
- 3 Give 10 examples for the application of computers in paper machine clothing.

Future of Papermaking and Paper Machine Clothing

With the current speed of technology development, it is hardly possible to predict the future of any industry. Nevertheless, some attempt is made in this chapter to predict some of the future trends in papermaking and paper machine clothing.

In the 1800s, a paper machine was producing a sheet of 5 feet wide at a speed of 100 fpm. Today, paper machines are close to 500 inches in width. The speed is at 7,000 fpm. Some pilot paper machines are already running at 8,000 fpm. It is expected that in a decade or so, the machine speeds will reach 10,000 fpm. The cost of a new machine can be over \$400 million. Parallel to the improvements in paper machine technology, paper machine clothing manufacturing also made giant steps within the last few decades.

With the introduction of the first synthetic forming fabric in 1957, the research and development efforts in paper machine clothing made a dramatic turn. Metal wires did not allow development of design varieties. Synthetic fabrics allowed more complicated designs and structures in forming, press and dryer fabrics. However, pressures and high temperatures still present a challenge for paper machine clothing manufacturers which will require new products to meet the demands. Demand for quality and uniformity will continue for all grades of paper and paperboard. Ever increasing recycled fiber content will require development of more sophisticated fabrics to solve the problems encountered in

forming, pressing and drying.

In 1995, the forest, wood and paper industry executives released the Agenda 2020 to set the future goals for these industries. The Agenda identifies six priority areas for research: sustainable forest management, environmental performance, energy performance, improved capital effectiveness, recycling and sensors and control [1].

The search will continue for new fabrics and processes that will improve paper quality and increase paper machine productivity. Paper machine manufacturers, papermakers, paper machine builders and universities need to continue working together to take the paper machine clothing and papermaking to new heights. Development of paper machine clothing with the input from paper manufacturers will be more successful. The key in paper machine clothing business will be to combine custom designed products with a good technical service.

9.1 Paper Machines and Papermaking

New concepts for forming, pressing and drying are expected to be developed in the future. Environmental issues will continue to be a concern that will affect the future directions in papermaking and paper machines. More rational use of energy and water will become necessary. More automation including fuzzy logic controls will be implemented.

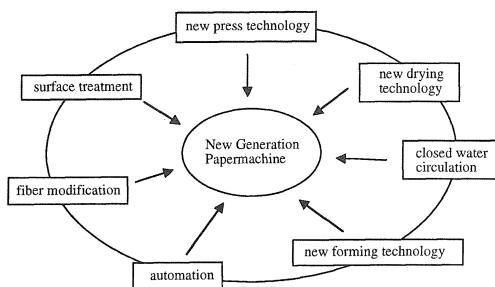


FIGURE 9.1. Factors that will affect the development of future paper machines [2].

Figure 9.1 shows the factors that will affect the future paper machine development. Layered headbox designs that will allow furnish layering will increase. The machine speeds and widths will continue to increase, the former being more likely. As the speed and tension of fabrics increase, more abrasion resistant materials will be developed for paper machine clothing. More hydraulic headboxes will be used in the future. Dilution control of basis weight rather than slice control allows for finer adjustments in basis weight profile without affecting fiber orientation [3]. Use of single fourdriniers will decline. Gap forming and twin wire machines will dominate the market for tissue and many paper grades. Adjustable blades are used in the latest generation of gap and blade formers. More multiple formers will be used for board grades. Former types will continue to evolve to meet the requirements of wider paper grades. Development work such as transfer belt technology will continue to have a fully supported web throughout the paper machine. High consistency forming will get more attention. Papermaking will move even more towards being science and engineering rather than an art and science.

Ever continuing progress of technology and lifestyles will necessitate new products with more demanding properties. More engineered papers including composite paper structures will be developed for high performance applications. Use of paper and paperboard products in industrial applications such as architecture and construction, medical technology, safety

and protection, etc., will increase. This will close the gap between nonwoven textiles and paper even further. As a result, it is expected that the percent of nonwood/nonfibrous materials will increase in paper and board products [4].

Recycled fiber consumption in North America was doubled between 1980-1993. The current recovery rate is 40%. Use of recycled fibers will continue to grow. The US Paper Industry set 50% paper recovery goal by 2000. The current recycling rate for Europe is 43% with Sweden leading at 50%. However, from an economical and technological point of view, there may be a limit to recycling increase in the future [5-8].

More two-sided coated grades will be made in bleached board. Liquid packaging (aseptic) grades are growing rapidly. Demand for recycled fiber in non-food packaging grades will increase. Use of chlorine as a bleaching agent will decrease. However, non-chlorine methods may reduce brightness levels. Use of oxygen bleaching agents such as ozone (O_3), hydrogen peroxide (H_2O_2) and others will increase for fine paper. Recycle fiber content will increase for brown paper. Departure from standard basis weight requirements will take place. Better printability of the sheet will be required. For tissue, increased wet strength (polyamide resins) will be required.

Automation in manufacturing of paper and paper machine clothing will continue to increase. This will require new sensors and new control strategies. Computer aided design and computer aided manufacturing will be utilized further.

9.2 Forming

Most forming fabric development efforts during the 1970's and 1980's focused on increasing drainage capacity and structural integrity including stability and life. These areas are of relatively lesser concern today because of improvements made in these areas. However, increasing machine speeds and widths will require even more stable and uniform

fabrics. Further stability in both MD and CD may be achieved by incorporating larger strands and new fabric designs. The use of polyamide yarns in the wear surfaces will become the norm. Increased use of abrasive calcium carbonate in alkaline sizing will continue to require excellent life potential and retention.

Product consistency and a generally improved product will continue to be searched. In general, fabrics with higher quality, longer life and lower cost will be developed. Improved consistency will result in repeatable and more dependable products.

Forming fabrics will need to offer greater levels of fiber support without sacrificing drainage or structural integrity. The extra support double-layer design family will continue to be actively used in the near future to meet the new demands. However, much more versatility with much less compromise is afforded to both the forming fabric design engineer and the papermaker through the use of lower caliper, three layer structures.

Forming fabrics will also need to be designed to run cleaner offering less affinity for contaminant attraction and fiber carryback. Anti-contaminant properties will be required even more as the use of recycled fiber is increased, especially in the brown grades. Because of recycled fiber and the changes it will make to wet-end chemistry, the forming fabric will be modified to help minimize contamination and plugging from these sources.

Increased demand from the printing and converting industries for smoother, stronger sheets with better printing surfaces will affect the forming fabric design and structure. In addition, the paper industry will have to produce these higher-quality sheets from lower-grade, less-expensive furnishes. Therefore, fabrics will have to be finer, giving more sheet support while improving their mechanical properties with regard to dimensional stability, wear resistance and ease of cleaning. These basically conflicting requirements can be met by multilayer designs and probably will be met by three-layer designs.

Paper machine speeds will continue to increase. The very short dewatering time then

necessitates forming fabrics with good drainage and high fiber support. In addition, the increased speed will require an ever-increasing need for fabric stability and resistance to stretching due to higher tensions. Easy fabric installation for less downtime will require development of pin and thin line coil seamed forming fabrics.

Paper weights will continue to decrease especially for fine paper while maintaining or improving current standards for opacity, smoothness, strength, printability, etc. In addition, other grades, while maintaining sheet weight, will need improvements in printability. All of these demands will require finer forming fabrics to enable the papermaker to obtain smoother sheet properties and more uniform fine distribution while achieving excellent formation. Finer fabrics will require multilayer designs to obtain necessary stability on the wider, high speed machines. Kraft papers will continue to see expansion in the multi-ply sheets and will also see finer fabrics to accommodate lighter weight plies and improvements in formation to improve ply bonding and printability.

Lower basis weights in tissue will be developed. Low basis weight in tissue can be achieved in different ways. The fibers can be oriented in the thickness direction (Z direction) during forming to give higher bulk. Another way to achieve low basis weight is multilayer through air dryer (TAD) fabrics with small pits. TAD will require higher temperature resistant fabrics with better hydrolysis resistance for greater efficiency and productivity.

9.3 Pressing

It is well known that better dewatering requires less energy and produces a superior product. Paper machine builders will continue working with paper machine clothing manufacturers to make further improvements in this area.

In the 1950's, press felts, which were composed of 100% wool fibers, had a life of approximately one week. Use of synthetic fibers

and new needling technologies increased the life drastically and revolutionized fabric designs. Double and triple layer base fabrics, twisted mono-filaments and stratified batt structures resulted in superior press fabric performance. More stable base structures and low batt-to-base ratio fabrics that will provide more uniform sheet profiles will be developed. Needling techniques will improve.

Usage of grooved belts and blind-drilled sleeves for shoe presses will increase on machines that are double-felted for kraft and other grades, as well as on single-felted operations.

The concept of no-open-draw is becoming a reality. No-open-draw will allow higher speeds by preventing sheet break. It is expected that shoe press type calender belts will replace soft nip calendering for board in the future.

Use of laminated press fabric structures will continue. Seamed fabrics utilizing a laminated top fabric are already being evaluated as are laminated types which are nonwovens with unique properties [9].

Alkaline papermaking for fine papers requires good water handling capacity which will require higher void volume structures. Couch consistency and pressability will be important. Felt filling due to carbonate and aluminum hydroxide will require coarser batt fibers with more open structures. Mechanical and chemical felt wear and shedding require high molecular weight batts and better chemical treatments.

Machine speeds and press loads will increase. Dwell times will be longer. Multiple shoe press sections and new Long Nip Press (LNP) sections will be utilized. To meet these challenges, single monofilament CD fabric constructions will be developed. Multilayer, multiple base fabric and pin seam constructions will increase. Chemical and wear resistant fiber content will increase. New and improved topical treatments will be developed for chemical and wear resistance.

To control the vibration and bounce in the fourth press, high caliper structures with improved mass uniformity are required. To improve sheet properties, pressure uniformity

will be improved. Press fabrics with densely woven base and fine denier fibers will be developed. New techniques to manufacture densely needled fabrics may prevent blowing.

Speeds will increase for bleached board pressing. There will be less double-felted pressing. High void volume designs will be used in first press positions. Smoother surface finish needling techniques will be developed.

For tissue and toweling, single-felted operations will increase. Softer pressure rolls will be used. Face-side high pressure showering and vacuum boxes will be utilized. Steam showers will be used in the nip.

Use of thermally stable nylon will increase dimensional stability in steam boxes/devronizers. High base-to-batt ratio designs will be developed for better compaction.

De-inking will become more important for newsprint. Higher machine speeds will require an extended nip press (ENP). Fourth presses will be common. Multi-nip or roll presses will likely be used for printing and writing grades. Third-position shoe presses are being used for various paper grades. Suction press rolls may be phased out and open draws between press and dryer sections may be eliminated [3].

Research and development work in impulse drying will continue. Combination of press loads and high temperatures provide better water removal with impulse drying. In this context, pressing and drying operations may be combined in the future. As a result, the paper machine length may be shorter.

9.4 Drying

Since the most difficult and expensive water removal takes place at the dryer section of the paper machine, there is room to make advances in the dryer section. Future dryer fabrics should maximize mass transfer and drying rate.

Use of high velocity, high temperature air impingement for more efficient drying will increase. A new concept in drying is Valmet's "Condebelt" dryer. In this system, in place of cylinders, two heated steel bands are used "between which the paper web is dried under

high pressure." It is claimed that this system increases dryer speed [3].

With increasing use of recycled fibers, there is a need to develop still better contaminant resistant dryer fabrics. Residual adhesives, pitch and ink from the recycled fiber, coating and size press solids can fill the dryer fabrics prematurely and affect drying efficiency, dryer fabric performance and fabric life. Better contaminant resistant fabrics will increase steam savings and fabric life, reduce downtime for cleaning and improve sheet quality.

Higher temperature, wear resistant dryer fabrics with good hydrolysis resistant will be developed. Less expensive yarn materials that will provide better hydrolysis resistance are in demand.

Low caliper, smooth surface monofilament fabrics will be developed. Properly shaped monofilaments both in MD and CD directions will be used to achieve this. Improvements in surface contact area will improve dryer efficiency. Minimal seam marking is required. Fabrics with less air carrying properties need to be developed.

Use of single run and single tier dryer configurations has increased in recent years. These configurations support the sheet continuously without open draws. In these machines, speed variations may occur between the top and bottom cylinders which may cause sheet stretching, slippage and break. The dryer fabric plays an important roll to reduce the amount of speed difference. There is still room for better dryer fabric stability and resistance to distortion and damage. The current move to single tier dryer systems will continue. Steam-heated, rotating, cast-iron cylinders will likely continue to be used. Closed transfers will become standard and fabric-driven dryers, with no ropes, will be used [3].

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Appendix A

Size-denier Relationships for Polyester and Nylon Monofilaments

POLYESTER MONOFILAMENT SIZE EQUIVALENCY CHART

Diameter Thousandths of an Inch (mil.)	Diameter Millimeters	Denier	Decitex	Yards/lb.	Meters/Kg.
.0039	.10	95	106	45,591	93,881
.004	.1016	100	112	44,289	89,243
.0043	.11	116	129	38,327	77,230
.0047	.12	139	154	32,078	64,638
.005	.1270	157	175	28,345	57,115
.0051	.13	163	182	27,245	54,899
.0055	.14	190	211	23,425	47,201
.0059	.15	219	243	20,357	41,020
.006	.1524	226	252	19,684	39,663
.0063	.16	250	277	17,854	35,977
.0067	.17	282	314	15,786	31,809
.007	.1788	308	343	14,461	29,140
.0071	.18	317	352	14,057	28,325
.0075	.19	354	393	12,598	25,385
.0079	.20	393	436	11,354	22,879
.008	.2032	403	448	11,072	22,310
.0083	.21	434	482	10,286	20,727
.0087	.22	476	529	9,362	18,865
.009	.2286	510	567	8,748	17,628
.0091	.23	521	579	8,557	17,243
.0094	.24	556	618	8,020	16,160
.0098	.25	605	672	7,378	14,867
.010	.2540	630	700	7,086	14,278
.0102	.26	655	728	6,811	13,724
.0106	.27	707	786	6,306	12,708
.011	.28	765	850	5,832	11,751
.0114	.29	818	909	5,452	10,987
.0118	.30	877	974	5,089	10,254
.012	.3048	907	1,008	4,921	9,915
.0122	.31	937	1,041	4,761	9,593
.0126	.32	1,000	1,111	4,463	8,994
.013	.33	1,064	1,183	4,193	8,449
.0134	.34	1,131	1,256	3,946	7,952
.0138	.35	1,199	1,333	3,721	7,497
.014	.3556	1,234	1,372	3,615	7,285
.0142	.36	1,270	1,411	3,514	7,081
.0146	.37	1,342	1,492	3,324	6,698
.015	.38	1,417	1,575	3,149	6,346
.0154	.39	1,494	1,660	2,988	6,020
.0157	.40	1,552	1,725	2,874	5,792
.016	.4064	1,612	1,792	2,768	5,577
.0161	.41	1,633	1,814	2,733	5,508
.0165	.42	1,715	1,905	2,602	5,244
.0169	.43	1,799	1,999	2,481	4,999
.017	.4318	1,820	2,023	2,452	4,940
.0173	.44	1,885	2,095	2,367	4,770
.0177	.45	1,973	2,193	2,261	4,557
.018	.4572	2,041	2,268	2,187	4,407
.0181	.46	2,063	2,293	2,163	4,358
.0185	.47	2,156	2,395	2,070	4,172
.0189	.48	2,250	2,500	1,983	3,997
.019	.4826	2,274	2,527	1,962	3,955
.0193	.49	2,346	2,607	1,902	3,833
.0197	.50	2,444	2,716	1,825	3,676

POLYESTER MONOFILAMENT SIZE EQUIVALENCY CHART

Diameter Thousandths of an Inch (mil.)	Diameter Millimeters	Denier	Decitex	Yards/lb.	Meters/Kg.
.020	.5080	2,520	2,800	1,771	3,567
.0201	.51	2,545	2,828	1,754	3,534
.0205	.52	2,647	2,941	1,686	3,397
.0209	.53	2,751	3,057	1,622	3,268
.021	.5334	2,778	3,087	1,606	3,237
.0213	.54	2,858	3,175	1,561	3,147
.0217	.55	2,966	3,296	1,504	3,032
.022	.56	3,060	3,400	1,458	2,939
.0224	.57	3,161	3,512	1,412	2,845
.0228	.58	3,274	3,638	1,363	2,746
.023	.5842	3,332	3,703	1,339	2,699
.0232	.59	3,390	3,767	1,316	2,652
.0236	.60	3,508	3,898	1,272	2,563
.024	.61	3,628	4,032	1,230	2,478
.0244	.62	3,750	4,167	1,190	2,398
.0248	.63	3,874	4,305	1,152	2,321
.025	.6350	3,937	4,375	1,133	2,284
.0252	.64	4,000	4,445	1,115	2,248
.0256	.65	4,128	4,587	1,081	2,178
.026	.66	4,258	4,732	1,048	2,112
.0264	.67	4,390	4,878	1,016	2,048
.0268	.68	4,524	5,027	986	1,988
.027	.6858	4,592	5,103	972	1,958
.0272	.69	4,660	5,178	957	1,929
.0276	.70	4,799	5,332	930	1,874
.028	.71	4,939	5,488	903	1,821
.0283	.72	5,045	5,606	844	1,782
.0287	.73	5,189	5,765	860	1,733
.029	.7366	5,298	5,887	842	1,697
.0291	.74	5,334	5,927	836	1,686
.0295	.75	5,482	6,091	814	1,640
.0299	.76	5,632	6,258	792	1,597
.030	.7620	5,670	6,300	787	1,586
.0303	.77	5,783	6,426	771	1,555
.0307	.78	5,937	6,596	751	1,515
.031	.7874	6,054	6,727	737	1,485
.0311	.79	6,093	6,770	732	1,476
.0315	.80	6,251	6,945	714	1,439
.0319	.81	6,410	7,123	696	1,403
.032	.8128	6,451	7,168	692	1,394
.0323	.82	6,572	7,303	679	1,368
.0326	.83	6,695	7,439	666	1,343
.033	.84	6,860	7,623	650	1,311
.0334	.85	7,028	7,808	635	1,279
.0338	.86	7,197	7,997	620	1,249
.034	.8636	7,282	8,092	613	1,235
.0342	.87	7,368	8,187	605	1,220
.0346	.88	7,542	8,380	591	1,192
.035	.89	7,717	8,575	578	1,165
.0354	.90	7,894	8,772	565	1,139
.0358	.91	8,074	8,971	552	1,114
.036	.9144	8,164	9,072	546	1,101
.0362	.92	8,255	9,173	540	1,089
.0366	.93	8,439	9,376	529	1,065
.037	.94	8,624	9,583	517	1,043
.0374	.95	8,812	9,791	506	1,020
.0377	.96	8,954	9,949	498	1,004
.038	.9652	9,097	10,108	490	988
.0381	.97	9,145	10,161	488	983
.0385	.98	9,338	10,375	478	963
.0389	.99	9,533	10,592	468	943
.039	.9906	9,582	10,646	465	938
.0393	1.00	9,730	10,811	458	924
.0397	1.01	9,929	11,032	449	905
.040	1.0160	10,080	11,200	442	892

NYLON MONOFILAMENT SIZE EQUIVALENCY CHART

Diameter Thousandths of an Inch (mil.)	Diameter Millimeters	Denier	Decitex	Yards/lb.	Meters/Kg.
.004	.1016	83	92	53,658	108,122
.0043	.11	96	106	46,436	93,569
.0047	.12	114	127	38,868	78,319
.005	.1270	130	144	34,341	69,198
.0051	.13	135	150	33,008	66,512
.0055	.14	157	174	28,381	57,188
.0059	.15	181	201	24,663	49,697
.006	.1524	187	208	23,848	48,054
.0063	.16	206	229	21,632	43,588
.0067	.17	233	259	19,126	38,538
.007	.1778	254	283	17,521	35,305
.0071	.18	262	291	17,031	34,317
.0075	.19	292	325	15,262	30,754
.0079	.20	324	360	13,756	27,719
.008	.2032	332	369	13,414	27,030
.0083	.21	358	398	12,462	25,112
.0087	.22	393	437	11,343	22,856
.009	.2286	421	468	10,599	21,357
.0091	.23	430	478	10,367	20,890
.0094	.24	459	510	9,716	19,578
.0098	.25	499	554	8,939	18,013
.010	.2540	520	577	8,585	17,299
.0102	.26	541	601	8,252	16,628
.0106	.27	584	649	7,641	15,396
.011	.28	629	699	7,095	14,297
.0114	.29	675	750	6,606	13,311
.0118	.30	724	804	6,165	12,424
.012	.3048	748	832	5,962	12,013
.0122	.31	773	859	5,768	11,623
.0126	.32	825	917	5,407	10,896
.013	.33	878	976	5,080	10,236
.0134	.34	933	1,037	4,781	9,634
.0138	.35	990	1,100	4,508	9,084
.014	.3556	1,019	1,132	4,380	8,826
.0142	.36	1,048	1,165	4,257	8,579
.0146	.37	1,108	1,231	4,027	8,115
.015	.38	1,170	1,300	3,815	7,688
.0154	.39	1,233	1,370	3,620	7,294
.0157	.40	1,281	1,424	3,483	7,018
.016	.4064	1,331	1,479	3,353	6,757
.0161	.41	1,347	1,497	3,312	6,673
.0165	.42	1,415	1,573	3,153	6,354
.0169	.43	1,485	1,650	3,005	6,057
.017	.4318	1,502	1,669	2,970	5,985
.0173	.44	1,556	1,729	2,868	5,780
.0177	.45	1,629	1,810	2,740	5,521
.018	.4572	1,684	1,872	2,649	5,339
.0181	.46	1,703	1,892	2,620	5,280
.0185	.47	1,779	1,977	2,508	5,054
.0189	.48	1,857	2,063	2,403	4,842
.019	.4826	1,877	2,085	2,378	4,792
.0193	.49	1,936	2,152	2,304	4,644
.0197	.50	2,018	2,242	2,212	4,457
.020	.5080	2,080	2,311	2,146	4,324
.0201	.51	2,100	2,334	2,125	4,281
.0205	.52	2,185	2,428	2,042	4,116
.0209	.53	2,271	2,523	1,965	3,960

NYLON MONOFILAMENT SIZE EQUIVALENCY CHART

Diameter Thousandths of an Inch (mil.)	Diameter Millimeters	Denier	Decitex	Yards/lb.	Meters/Kg.
.021	.5334	2,293	2,548	1,946	3,922
.0213	.54	2,359	2,621	1,892	3,813
.0217	.55	2,448	2,720	1,823	3,673
.022	.56	2,516	2,796	1,773	3,574
.0224	.57	2,609	2,899	1,711	3,447
.0228	.58	2,703	3,003	1,651	3,327
.023	.5842	2,750	3,056	1,622	3,270
.0232	.59	2,798	3,109	1,595	3,214
.0236	.60	2,896	3,217	1,541	3,106
.024	.61	2,995	3,328	1,490	3,003
.0244	.62	3,095	3,439	1,442	2,905
.0248	.63	3,198	3,553	1,395	2,812
.025	.6350	3,250	3,611	1,373	2,767
.0252	.64	3,302	3,669	1,351	2,724
.0256	.65	3,407	3,786	1,310	2,639
.026	.66	3,515	3,905	1,270	2,559
.0264	.67	3,624	4,026	1,231	2,482
.0268	.68	3,734	4,149	1,195	2,408
.027	.6858	3,790	4,212	1,177	2,373
.0272	.69	3,847	4,274	1,160	2,338
.0276	.70	3,961	4,401	1,127	2,270
.028	.71	4,076	4,529	1,095	2,206
.0283	.72	4,164	4,627	1,071	2,106
.0287	.73	4,283	4,759	1,042	2,100
.029	.7366	4,373	4,859	1,020	2,057
.0291	.74	4,403	4,892	1,013	2,042
.0295	.75	4,525	5,028	986	1,987
.0299	.76	4,648	5,165	960	1,935
.030	.7620	4,680	5,200	953	1,922
.0303	.77	4,774	5,304	935	1,884
.0307	.78	4,900	5,445	910	1,835
.031	.7874	4,997	5,552	893	1,800
.0311	.79	5,029	5,588	887	1,788
.0315	.80	5,159	5,733	865	1,743
.0319	.81	5,291	5,879	843	1,700
.032	.8128	5,324	5,916	838	1,689
.0323	.82	5,425	6,027	822	1,658
.0326	.83	5,526	6,140	807	1,627
.033	.84	5,662	6,292	788	1,588
.0334	.85	5,800	6,445	769	1,550
.0338	.86	5,940	6,600	751	1,514
.034	.8636	6,011	6,679	742	1,496
.0342	.87	6,082	6,757	734	1,479
.0346	.88	6,225	6,916	717	1,445
.035	.89	6,370	7,077	700	1,412
.0354	.90	6,516	7,240	685	1,380
.0358	.91	6,664	7,405	669	1,349
.036	.9144	6,739	7,488	662	1,334
.0362	.92	6,814	7,571	655	1,320
.0366	.93	6,965	7,739	640	1,291
.037	.94	7,118	7,909	627	1,263
.0374	.95	7,273	8,081	613	1,236
.0377	.96	7,390	8,211	604	1,217
.038	.9652	7,508	8,343	594	1,198
.0381	.97	7,548	8,387	591	1,191
.0385	.98	7,707	8,564	579	1,167
.0389	.99	7,868	8,742	567	1,143
.039	.9906	7,909	8,788	564	1,137
.0393	1.00	8,031	8,923	555	1,120
.0397	1.01	8,195	9,106	544	1,097
.040	1.016	8,320	9,244	536	1,081
.045	1.143	10,530	11,700	423	854
.050	1.270	13,000	14,444	343	691
.055	1.397	15,730	17,477	283	571
.060	1.524	18,720	20,800	328	480
.065	1.651	21,970	24,411	203	409
.070	1.778	25,480	28,311	175	353
.075	1.905	29,250	32,500	152	307
.080	3.032	33,280	36,977	134	270



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Appendix B

Forming Section Troubleshooting Guide

<i>Symptom</i>	<i>Possible Causes/Check Points/Possible Solutions</i>
Barring	—Check conditions of rolls (vibration) —Check seals of the couch roll —Check for variation on all machine elements —Check for loose bearing houses —Compare mark to felt pattern —Check for pressure pulses • headbox • pressure screen • stock delivery system
Cockling	—Poor formation —Water jumps—stock jump —Water circulation from suction couch roll —Reduce refining —Pulsating on the stock system —Rewetting of the fabric from showers —Water spots from dandy roll or couch press
Couch Shadow Marking	—Check drainage—may be flooding couch —Check vacuums—increase drainage on table —Check drag to jet ratio —Check fabric tension—may be too low —Check for excessive fabric wear —Review fabric drainage capacity—may need to increase fabric drainage —Freeness too low
Cracks	—Check machine deckles —Check trim —Rewetting on edges

<i>Symptom</i>	<i>Possible Causes/Check Points/Possible Solutions</i>
Crushing	—Sheet running wet into pressing elements —Check draws (speed differences) —Check couch press —Check dandy roll —Dandy out of balance —Sheet too wet at dandy —Dandy set too hard —Consistency too high —Fabric plugged —Suction couch roll showers not set correctly
Curl	—Fiber alignment —Two sidedness —Refining —Draw —Edge dewatering —Increase long fiber —Reduce freeness of stock —Retention aid uneven —Fabric dewatering uneven —Check headbox slice opening
Dandy Roll Marks	—Plugged dandy shower —Damaged dandy roll —Dandy not level —Check alignment of dandy —Sheet coming into dandy, too wet —Sheet too dry—picking sheet —Check vacuum levels of suction boxes
Edge Breaks	—Trim squirt plugged—not cutting clean —Recouching of sheet and edge trim in pick nip —Foil, couch roll—trim too small —Dirt build up on bottom lip —Dirt build up on fabric edge —Stock build up on deckles —Poor dewatering of edge area —Rewetting at couch —Fabric damaged —Stock build up on foils or suction boxes —Fabric edge damage from suction box sealing —Deckles touching fabric —Stock build up on return rolls
Guiding	—High vacuum on suction boxes —Improper guide palm sensitivity —Non-uniform fabric tension—front to back —Insufficient wrap on guide roll

<i>Symptom</i>	<i>Possible Causes/Check Points/Possible Solutions</i>
	—Diagonal of drilled boxes —Check drive sync —Check for fabric slippage —Check doctor blades —Check roll bearings —Check fabric tension —Check draw —Check suction box vacuum and surface —Check for unbalance rolls —Check guide movement
Moisture Profile Uneven	—Uneven head box slice caliper —Stock jet speed difference in tubes —Stock speed difference —Uneven plates in headbox —Check rectifier roll —Deflection of breast roll —Check foils for unevenness—level side to side —Check foils for wear —Uneven fabric tension —Uneven shower water supply —Check for plugged suction rolls
Pin Holes	—Check for air in the system —Check for excessive drainage—stapling —Change table elements—too slow drainage —May need to close up fabric —Check vacuum of suction boxes—too high —Check for thin spots in sheet —Poor sheet release
Ply Bonding/Delamination	—Increase refining —Improve formation —Decrease internal sizing —Increase long fiber content —Increase strength additives
Poor Drainage	—Check if freeness is too high —Check if stock temperature is too low —Check pH —Check retention aids used —Check if refining is excessive —Check for air in stock —Check fabric for excessive wear —Check machine elements for wear —Check table alignment —Check fabric tension —Check for dirty fabrics

<i>Symptom</i>	<i>Possible Causes/Check Points/Possible Solutions</i>
	—Check for air leaks —Sheet sealing
Poor Formation	—Too high fabric oscillation—air —Vibration of table elements —Fabric slippage —Check for excessive fabric wear —High friction at suction boxes • vacuum too high • dirt • sharp edges —Increase fabric tension —Check for fabric deflection —Check for vacuum pulses —Check rush/drag—impingement—elements
Release	—Check for sheet stapling —Trim squirts not cutting edge —Check fabric for excessive drainage —Check fabric for excessive sheet side wear —Couch vacuum too high —Pickup vacuum too low —Pickup felt plugged—add water to pickup felt —Check fabric tension —Check draws —Check for wrinkles/ridges
Ridges/Wrinkles of Forming Fabrics	—Check machine alignment —Excessive table shake —Check high pressure shower nozzles —Check for deposits on table elements —Check for damaged foils—table elements —Check operating tension —Check pressure of high pressure showers —Check rolls and doctor blades for excessive wear —Check rolls and doctor blades for alignment, deflection and stability —Check shower flows and pressures —Check drive ratios —Check guide system, pressure and air supply —Check for wrong geometry of fabric oscillation
Sheet Wrinkles	—Too high stock speed in head box tube system —Base plates uneven —Plugged tubes —Slice uneven —Check for damaged slice —Check breast roll shower —Check for worn shower nozzles

<i>Symptom</i>	<i>Possible Causes/Check Points/Possible Solutions</i>
	—Check for breast roll alignment —Check for build up on breast roll —Deckle setting incorrect —Dirty fabric —Worn fabric —Waves in fabrics
Speed Variation on Fabrics	—Check for fabric slippage —Check for excessive friction —Check table elements —Check bearings —Check machine alignment —Check for drive overload—drag load —Check suction box vacuum—too high —Check for roll surfaces—too smooth —Check for slot width being too wide —Check for vacuum variation —Check for couch stability —Check for frame instability —Check for couch roll cover hardness variation —Check for roll balance
Streaky Formation	—Check headbox slice opening —Check for stock build up on slice —Check for slice distortion —Check headbox —Check rectifier roll —Check for wear streaks in fabric —Worn out slice —Check alignment of table —Check if fabric is dirty —Dirty dandy roll
Wire Mark	—Check refining —Check vacuum—reduce if possible —Check fabric tension—increase if possible —Check drag to jet ratio —Check for excessive fabric drainage —Reduce loading on dandy-couch press pickup felt —Check fabric for excessive sheet side wear —Different fabric may be needed



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Appendix C

Press Section Troubleshooting Guide

<i>Symptom</i>	<i>Possible Causes/Check Points/Possible Solutions</i>
Poor Pickup	—Increase felt water just ahead of the pickup zone —Check weight profile, especially if the pickup problem is not confined to the edges —Speed differential: Are loads and amperages normal? —“Wire” tension—try to increase—below 20 is too low (25 pli on suction breast roll, SBR) —Felt tension: Is it normal? Try slack, then tight. —Holes in the sheet—Is forming fabric contaminated? Check for fibers in spray as felt and forming fabric separate. Lots of fibers indicate poor pickup, not contamination. —Sheet stapled to forming fabric: Lower vacuum at SBR; check suction box position. —Lower felt pickup point (into forming fabric)
Crushing	—Decrease felt water content —Increase vacuum (check vacuum) —Clean felt —Inspect for plugged uhle box, suction roll holes, blind-drilled hole, etc. —Lower machine speed while high pressure showering —Check for felt surface damage —Check for roll cover slippage
Blistering	—Check for uniform Yankee coating (check spray boom coverage) —Check for uneven loading/wear on Yankee doctor blade —Hot edges (add water to the edges) —Lower steam pressure in Yankee, increase hood temperature —Check Yankee differential pressure (high differential creates hot spots resulting in blistering. Good number is 12–16 psi). —Check thermal topography of the Yankee

<i>Symptom</i>	<i>Possible Causes/Check Points/Possible Solutions</i>
Sheet Following	—Check felt moisture ratio —Check for uniform Yankee coating. If uneven, check Yankee doctor blade and spray bottom —Check for filled, compacted, smooth areas in felt and clean accordingly —Check suction pressure roll box position (throw off should be 8°–10° from the exiting felt tangent) and seal strips —Lower felt water—has pressure roll crown been changed —Check pressure roll loading and increase
Felt Bagginess	—Check felt tension (fabric slippage) —Check for worn uhle box strips, pickup shoes/bars and their proper lubrication —Check for temperature differentials across the face of the fabric —Check for Yankee hood leaks on the surface of the fabric
Sheet Moisture Profile Problems	—Determine where problem was seen first —Check for uneven basis weight profile —Check for damaged forming fabric —Measure felt moisture—does problem correlate? —Is the felt skewed? —Clean felt —Uneven shower application—uneven cleaning —Check air pressure on the seal strips —Are seal strips loaded evenly? —Plugged/worn uhle box —Check hood profiling dampers —Check for roll plugging —Check for roll cover slippage —Is the sheet profiler operating correctly? —Doctor loading on the pressure rolls or main rolls —Doctor loading on the Yankee —Check spray boom for nozzle plugging
Sheet Stealing (to bottom felt)	—Reduce loading at the main press —Reduce bottom felt tension —Vary water flow (usually lower) —Check uhle box vacuum —Rotate the main press suction box toward the wet end —Check air pressure on the seal strips —Are seal strips loaded evenly? —Is bottom felt filled? If so, clean. —Lightly brush the bottom felt surface —Turn bottom felt inside out

Tension Force Calculations of Dryer Fabrics

INTRODUCTION

To enable a dryer fabric to perfectly fulfil its role, it is necessary that it is properly tightened.

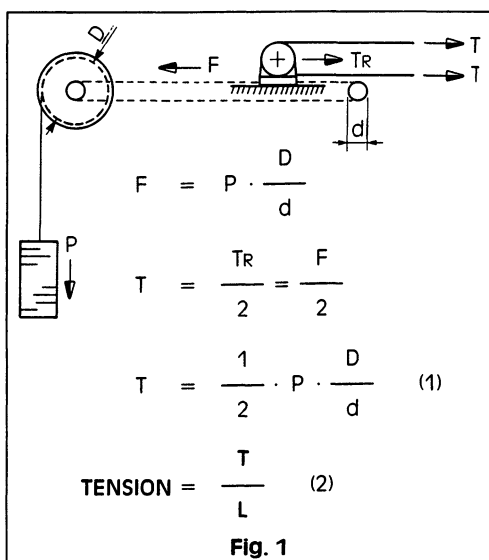
This tension is generally expressed in kilograms per centimeter (kg/cm) of width of fabric and ideally it should be between 1.5 and 1.8 kg/cm.

Below 1.5 kg/cm, the contact between the paper and the cylinder is negatively affected and consequently, the drying capacity is diminished.

Above 1.8 kg/cm, excessive mechanical tension could lead to excessive deflection of the rolls and can cause premature rupture of fabric and/or seam.

1. GENERAL SITUATION

On most paper machines, the tension regulation systems of dryer fabrics (if they exist) are schematized as follows:



Force "F" which acts on the stretch roll depends on force "P", shown here by weights but can equally be a hydraulic system or compressed air or be exerted by a motor.

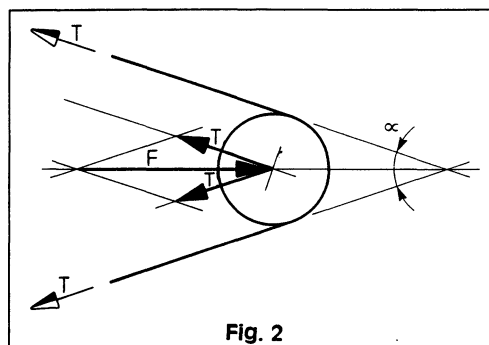
Dividing force "T" by the width of the fabric "L" presents the effective tension.

2. PARTICULAR CASES

The preceding formulas can be used with good approximation when the wrap on the stretch roll is approximately 180°C and the force "F" acting on it is parallel to the reaction force "TR" of the stretch roll.

When these two conditions are not simultaneously met, it is important to know that in establishing the formula (1) above, the cosine values of the intervening angles with the vectorial breakdown should be used.

2.1 The movement of the stretch roll follows the bisector of the angle formed by the fabric



F = force acting on the stretch roll in equilibrium with TR

TR = total reaction force of the fabric

After breaking down the vectorial component, we obtain:

$$T = \frac{F}{2} \cdot \frac{1}{\cos \frac{\alpha}{2}}$$

2.2 The movement of the stretch roll occurs parallel to one of the edges of the fabric

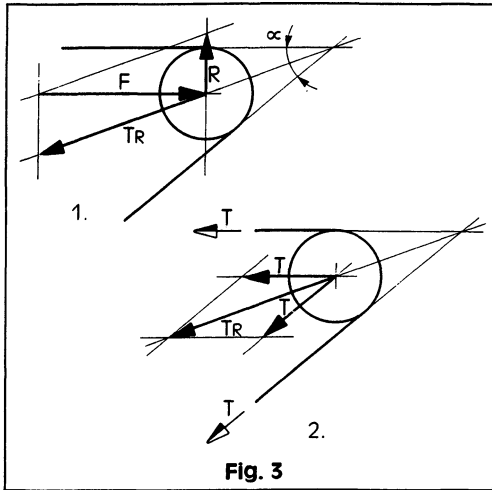


Fig. 3

The total reaction force "TR" of the fabric only balances the acting force "F" with the aid of the counteracting force "R" which represents the reaction of the frame of the machine.

By breaking down force "F" vectorially in Pos. 1, we obtain:

$$T_R = F \cdot \frac{1}{\cos \frac{\alpha}{2}}$$

The decomposition of "TR" on Pos. 2 gives:

$$T = \frac{T_R}{2} \cdot \frac{1}{\cos \frac{\alpha}{2}}$$

Finally:

$$T = \frac{F}{2} \cdot \frac{1}{\cos^2 \frac{\alpha}{2}}$$

2.3 The movement of the stretch roll can occur in any direction in relation to the direction of the acting forces

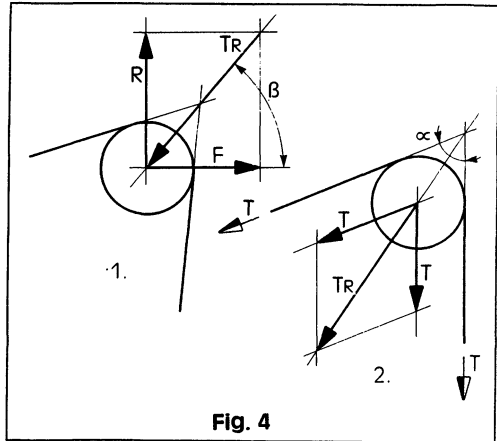


Fig. 4

F = Force acting on the stretch roll
R = Counteracting force of the machine frame
TR = Total counteracting force on the dryer fabric

The vectorial breakdown of "F" on Pos. 1 produces:

$$T_R = F \cdot \frac{1}{\cos \beta}$$

The breakdown of "TR" on Pos. 2 shows:

$$T = \frac{T_R}{2} \cdot \frac{1}{\cos \frac{\alpha}{2}}$$

Finally:

$$T = \frac{F}{2} \cdot \frac{1}{\cos \beta \cdot \cos \frac{\alpha}{2}}$$

2.4 Diagram of the values of

$$\frac{1}{\cos \frac{\alpha}{2}} \text{ and of } \frac{1}{\cos^2 \frac{\alpha}{2}} \text{ as}$$

a function of the angle "α"

Depending on the machines, the angle "α" can be strongly variable and can reach sometimes relatively high values when, for example, a dryer fabric has been shortened as a result of a new seam.

Another frequent example: a change in the return run of the fabric.

The diagram shows us that for "α" = 60°, the over-tension on the fabric compared to the classical case surpasses 30% in case 2.2 (relatively frequent) and even more than 15% in case 2.1.

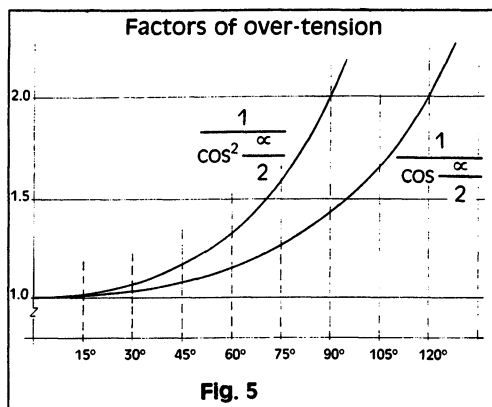


Fig. 5

3. SOME EXAMPLES OF THE CALCULATIONS OF "F" AND "T"

3.1 Bottom position of the section

Effect of the weight of the stretch roll

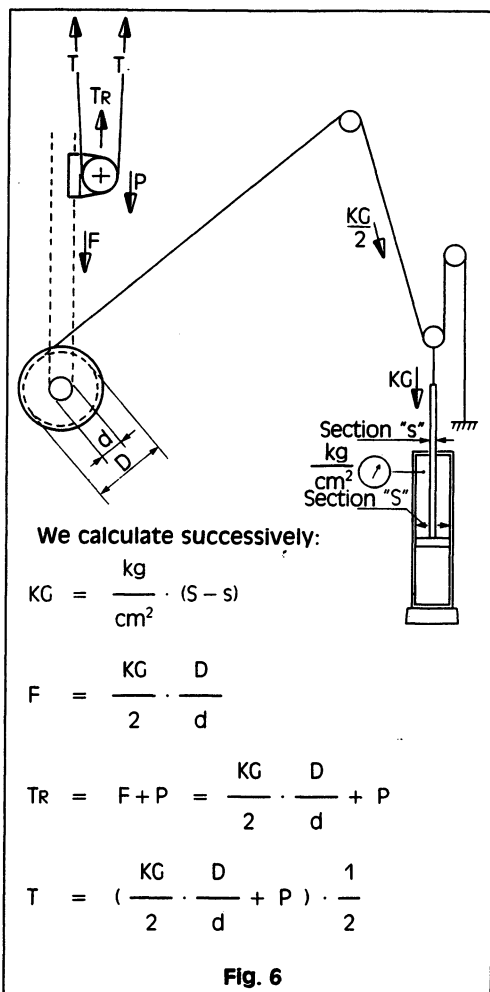
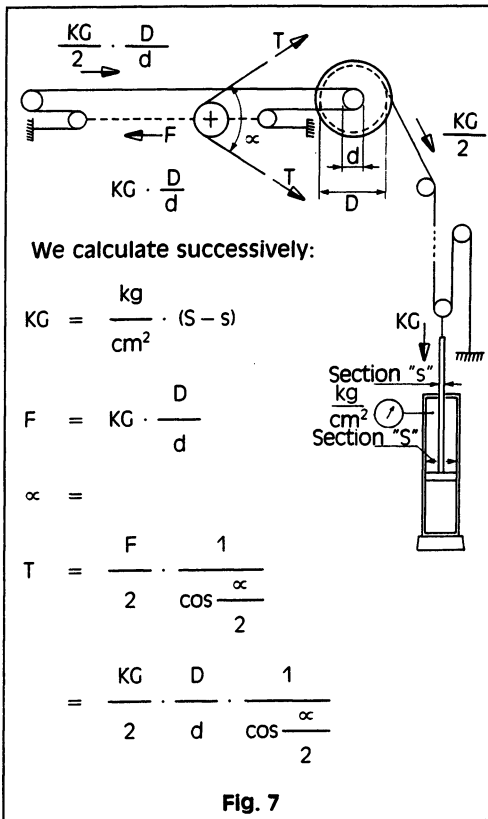
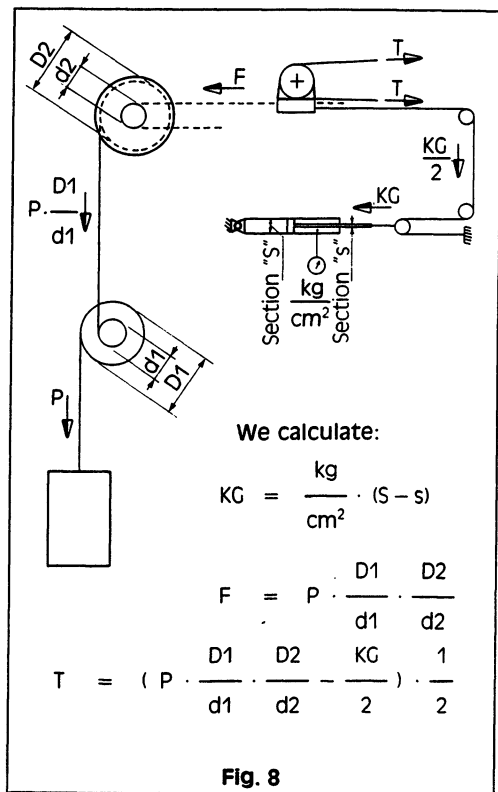


Fig. 6

3.2 Influence of a second system of pulleys

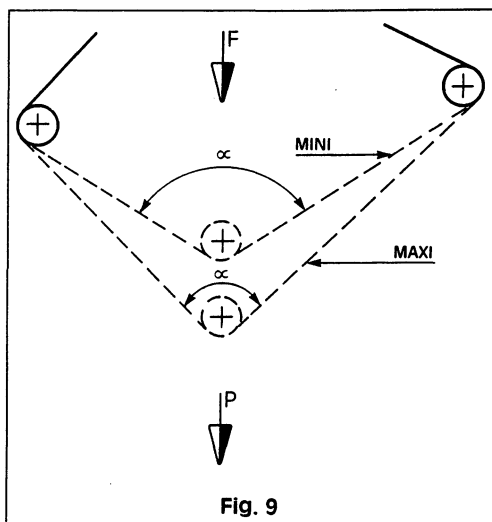


3.3 Influence of a pneumatic returning stretch roll



3.4 Influence of the angle " α "

The movement of the stretch roll occurs according to the bisector of the angle " α " formed by the fabric



In this machine, the possible movement of the stretch roll is not very important.

The angle " α " varies only slightly between the maximum and minimum positions of the stretch roll.

$$90^\circ \leq \alpha \leq 115^\circ$$

As per 2.1 and 3.1 above, the force on the fabric is calculated as follows:

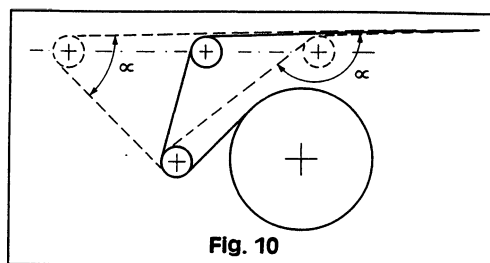
$$T = (F + P) \cdot \frac{1}{2} \cdot \frac{1}{\cos \frac{\alpha}{2}}$$

The factor of over-tension shall be therefore situated between the values:

$$1.41 \leq \frac{1}{\cos \frac{\alpha}{2}} < 1.86$$

3.5 Influence of the angle " α "

The movement of the stretch roll occurs parallel to one side of the fabric



On this machine, it was calculated that the angle " α " for the maximum and minimum positions of the stretch roll varies between the extremities:

$$46^\circ \leq \alpha \leq 143^\circ$$

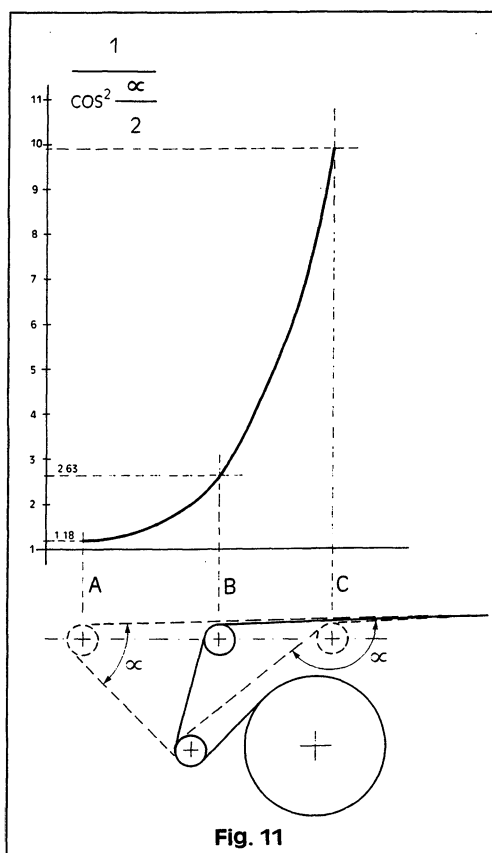
As per 2.2 above, the tension of the fabric is calculated as follows:

$$T = \frac{F}{2} \cdot \frac{1}{\cos^2 \frac{\alpha}{2}}$$

The over-tension factor is therefore between the values:

$$1.18 \leq \frac{1}{\cos^2 \frac{\alpha}{2}} \leq 9.93$$

To regulate correctly the manometric pression required, it is vital to take into consideration the position of the stretch roll on the frame.



3.5.1

In this machine, not considering the over-tension factor, a manometric pressure of 3.5 kg/cm^2 corresponds to a fabric tension of 1.6 kg/cm .

3.5.2

In fact, the extreme position "A" of the stretch roll would require a manometric pressure of:

$$\frac{3.5}{1.18} = 2.97 \text{ kg/cm}^2$$

in order to generate the fabric tension of 1.6 kg/cm .

3.5.3

Similarly, in position "B", the manometric pressure should be:

$$\frac{3.5}{2.63} = 1.33 \text{ kg/cm}^2$$

in order to generate a same tension of 1.6 kg/cm .

Ing. Marcel SIQUET
Applications Engineer
January 1990



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Appendix E

List of Testing Laboratories/ Research Facilities

TAPPI does not sanction the companies cited in this list. TAPPI provides this information as a service to its members and the general public. This information was obtained from a survey of industry test laboratories and research facilities. To determine the services they offer, please contact individual laboratories. To be included in the current test laboratories list, please call (770) 209-7208.

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Marietta, GA 30066
Contact Person: Bob Dunning or Phil Rogers
Tel: (404)423-1400

1218 Donaldson Rd.
Greenville, SC 29605
Contact Person: Paul West, Manager
Tel: (803)299-0525

108-A Castle Dr.
Madison, AL 35758
Contact Person: John Cooley, Manager

ARTHUR D. LITTLE, INC.
25 Acorn Park
Cambridge, MA 02140
Contact Person: Dr. Judith Harris
Tel: (617)864-5770

AUBURN UNIVERSITY
Pulp and Paper Research and Education Center (PPREC)
242 Ross Hall
Auburn, AL 36849-5128
Director: Dr. Ronald D. Neuman
Tel: (205)826-5223

BADGER LABORATORIES & ENGINEERING COMPANY, INC.
1110 South Oneida Street
Appleton, WI 54915
President: Arthur B. Kaplan
Tel: (414)739-9213
(800)242-3556

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505 King Avenue

Columbus, OH 43201
Tel: (614)424-6424

CHICAGO TESTING LABORATORY, INC.
3360 Commerical Avenue
Northbrook, IL 60062
President: Conway C. Burton
Tel: (312)498-6400

CONTAINER TESTING LABORATORY, INC.
607 Fayette Ave.
Mamaroneck, NY 10543
Tel: (914)381-2600
Fax: (914)381-0143

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Canada
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Fax: (819)372-9938

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Black Canyon Stage One
Phoenix, AZ 85029
Tel: (602)465-7356

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Annacis Island
New Westminster, BC V3M 5R1
Canada
Pulping & Bleaching: Randy Lowe and Tadas Macas
Fibre Testing: Wendy McAlpine
Analytical: Terry Peel
Tel: (604)526-4221
Fax: (604)526-1898

U.S. Mailing Address:
P.O. Box 952
Point Roberts, WA 98281

FOREST SERVICE
Forest Products Laboratory
1 Gifford Pinchot Dr.
Madison, WI 53705-2398
Director: John R. Erickson
Tel: (608)231-9200
Fax: (608)231-9592

GALBRAITH LABORATORIES, INC.
2323 Sycamore Drive
Knoxville, TN 37921
Tel: (615)546-1335

GORHAM INTERNATIONAL, INC.
P. O. Box 8
Gorham, ME 04038
Contact Person: Alfred Belisle
Tel: (207)892-2216

HERTY FOUNDATION
P. O. Box 7798
Garden City, GA 31418
Director: Dr. Michael J. Kocurek
Tel: (912)964-5541
Fax: (912)964-5614

INDUSTRIAL TESTING AND RESEARCH LABORATORY
P. O. Box 43643
650 Great S.W. Pkwy.
Atlanta, GA 30336
Contact Person: Roberta Louise Johnson
Tel: (404)344-7883

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Atlanta, GA 30318
President: James Ferris

Tel: (404)853-9500
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3480 University Street
Montreal, PQ H3A 2A7
Canada
Coordinator: M. R. Kamal or L. St.
Pierre
Tel: (514)398-4262

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TECHNOLOGICAL
UNIVERSITY**

Institute of Wood Research
Houghton, MI 49931
Director: Dr. W. E. Frayer
Tel: (906)487-2464

**NORTH CAROLINA STATE
UNIVERSITY**

Pulp and Paper Laboratory
Box 8005
Raleigh, NC 27695-8005
Department Head: Dr. Richard J.
Thomas
Tel: (919)515-2888
Fax: (919)515-7231
Wood Products Laboratory
Box 8005
Raleigh, NC 27695-8005
Department Head: Dr. Richard J.
Thomas
Tel: (919)515-2881
Fax: (919)515-7231

**PRO-PACK TESTING
LABORATORY, INC.**

15 North Florida
Belleville, IL 62221
Contact Person: Manuel Rosa
Tel: (618)277-1160
Fax: (618)277-1163

**PULP AND PAPER RESEARCH
INSTITUTE OF CANADA**

570 St. John's Boulevard
Pointe, PQ H9R 3J9
Canada
Contact Person: Wilfried Johasch
Tel: (514)630-4100
Fax: (514)630-4134

**SUNY COLLEGE OF
ENVIRONMENTAL SCIENCE
AND FORESTRY**

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Institute
Syracuse, NY 13210
Director: Dr. Leland R. Schroeder
Tel: (315)470-6502

Paper Science & Engineering Pilot
Plant
Syracuse, NY 13210
Director: Dr. William Holtzman
Tel: (315)470-6506
Fax: (315)470-6779

TECHNIDYNE CORPORATION

100 Quality Ave.
New Albany, IN 47150-2272
Contact Persons: Thomas Crawford
and Pat Robertson
Tel: (812)948-2884
Fax: (812)945-6847

**UNIVERSITY OF
CALIFORNIA, BERKELEY**

Forest Products Laboratory
1301 South 46th Street
Richmond, CA 94804
Director: Dr. Frank C. Beall
Tel: (415)231-9452

**UNIVERSITY OF LOWELL
RESEARCH FOUNDATION**

450 Aiken Street
Lowell, MA 01854
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Appendix F

Units and Conversion Factors

1. U.S. CUSTOMARY UNITS

Length	1 foot = 12 inches 1 yard = 3 feet 1 mile = 1,760 yards = 5,280 feet 1 inch = 1,000 mils
Area	1 square foot = 144 square inches 1 square yard = 9 square feet 1 acre = 43,560 square feet 1 square mile = 640 acres
Volume	1 cubic yard = 27 cubic feet 1 cubic foot = 1,728 cubic inches 1 gallon = 231 cubic inches
	liquid or fluid measures
	1 pint = 4 gills = 16 ounces 1 quart = 2 pints 1 gallon = 4 quarts 1 cubic foot = 7.4805 gallon
	dry measures
	1 quart = 2 pints 1 peck = 8 quarts 1 bushel = 4 pecks
Weights	1 ounce = 16 drams = 437.5 grams 1 pound = 16 ounces = 7,000 grains 1 ton (US) = 2,000 lb
Pressure	1 atmosphere = 14.7 psi

2. SI PREFIXES

<i>multiplication factors</i>	<i>prefix</i>	<i>SI symbol</i>
1 000 000 000 000 000 000 = 10^{18}	exa	E
1 000 000 000 000 000 = 10^{15}	pecta	P
1 000 000 000 000 = 10^{12}	tera	T
1 000 000 000 = 10^9	giga	G
1 000 000 = 10^6	mega	M
1 000 = 10^3	kilo	k
100 = 10^2	hecto	h
10 = 10^1	deka	da
0.1 = 10^{-1}	deci	d
0.01 = 10^{-2}	centi	c
0.001 = 10^{-3}	milli	m
0.000 001 = 10^{-6}	micro	μ
0.000 000 001 = 10^{-9}	nano	n
0.000 000 000 001 = 10^{-12}	pico	p
0.000 000 000 000 001 = 10^{-15}	femto	f
0.000 000 000 000 000 001 = 10^{-18}	atto	a

3. SI UNITS

<i>quantity</i>	<i>unit</i>	<i>SI symbol</i>	<i>Formula</i>
Base Units			
length	meter	m	
mass	kilogram	kg	
time	second	s	
electric current	ampere	A	
thermodynamic temperature	Kelvin	K	
amount of substance	mole	mol	
luminous intensity	candela	cd	
Derived Units			
acceleration	meter per second square		m/s^2
area	square meter		m^2
density	kilogram per cubic meter		kg/m^3
energy	joule	J	$\text{N}\cdot\text{m}$
force	Newton	N	$\text{kg}\cdot\text{m/s}^2$
frequency	hertz	Hz	$1/\text{s}$
power	Watt	W	J/s
pressure	Pascal	Pa	N/m^2
velocity	meter per second		m/s
voltage	volt	V	W/A
volume	cubic meter		m^3
work	joule	J	$\text{N}\cdot\text{m}$

1 meter = 100 cm = 1,000 millimeter

1 angstrom = 10^{-8} centimeter

1 ton = 1,000 kg
 1 kilogram = 1,000 gram
 1 kilogram = 9.807 Newton
 1 Newton = 100,000 dyn
 1 liter = 100 centiliter = 1,000 milliliter = 1,000 cubic centimeter
 1 cubic meter = 1,000,000 cubic centimeter
 1 atmosphere = 1.01×10^5 Pascal
 $1 \text{ kg/cm}^2 = 0.9678 \text{ atmosphere}$
 $1 \text{ bar} = 10^6 \text{ dynes/cm}^2 = 10^5 \text{ Newton/m}^2$
 1 Pascal = 1 Newton/m²
 1 cm of mercury (0° C) = 13.60 cm of water (4° C)
 1 kWh = 3.6×10^6 Joules
 1 Watts = 1 Joules/second = 1.34×10^{-3} HP

4. CONVERSION BETWEEN SI AND U.S. CUSTOMARY UNITS

1 miles = 1.609 kilometer
 1 yard = 0.9144 m
 1 inch = 25.4 mm
 $1 \text{ m}^2 = 1.196 \text{ square yard}$
 1 acre = 4,047 m²
 1 cubic inch = 16.39 cubic centimeter
 1 lb = 0.453 59 kg = 453.59 g
 1 quart = 0.9464 liters
 1 ounce = 28.35 grams
 1 ounce (liquid) = 0.02957 liters
 $1 \text{ kg/m}^2 = 9.807 \text{ Pascal} = 0.001422 \text{ pounds/square inch}$
 $1 \text{ lb/square foot} = 4.882 \text{ kg/m}^2$
 1 ton (metric) = 2,205 lbs
 1 BTU = 1,054 Joules
 1 Joule = 0.7376 ft-lbs
 1 HP = 550 ft-lb/second
 1 Newton = 0.2248 lbs
 $1 \text{ m/s}^2 = 3.281 \text{ feet/s}^2$
 $1 \text{ cfm} = 18.29 \text{ m}^3/\text{m}^2/\text{h}$

Metric to US Customary Conversion Factors

<i>to get</i>	<i>multiply</i>	<i>by</i>
inches	centimeters	0.3937007874
feet	meters	3.280839895
yards	meters	1.093613298
miles	kilometers	0.6213711922
ounces	grams	0.03527396195
pounds	kilograms	2.204622622
gallons (liquid)	liters	0.2641720524
fluid ounces	mililiters (cc)	0.03381402270

<i>to get</i>	<i>multiply</i>	<i>by</i>
square inches	centimeter squares	0.1550003100
square feet	meter squares	10.76391042
square yards	meter squares	1.195990046
cubic inches	milliliters (cc)	0.06102374409
cubic feet	cubic meters	35.31466672
cubic yards	cubic meters	1.307950619

US Customary to Metric

<i>to get</i>	<i>multiply</i>	<i>by</i>
microns	mils	25.4
centimeters	inches	2.54
meters	feet	0.3048
meters	yards	0.9144
kilometers	miles	1.609344
grams	ounces	28.34952313
kilograms	pounds	0.45359237
liters	gallons (liquid)	3.785411784
milliliters (cc)	fluid ounces	29.57352956
square centimeters	square inches	6.4516
square meters	square feet	0.09290304
square meters	square yards	0.83612736
milliliters (cc)	cubic inches	16.387064
cubic meters	cubic feet	0.02831684659
cubic meters	cubic yards	0.764554858

5. TEMPERATURE CONVERSION

$$^{\circ}\text{F} = 9/5 (^{\circ}\text{C}) + 32$$

$$^{\circ}\text{C} = 5/9 [(^{\circ}\text{F}) - 32]$$

$$^{\circ}\text{K (Kelvin)} = ^{\circ}\text{C} + 273$$

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